
Pre-requisites

This project utilizes modern Python technologies and Libraries for resume analysis, data processing, machine learning, skill extraction, and web-based deployment. The following tools and libraries are required for development and execution.

Python

Python is the primary programming language. It is used for implementing machine learning models, data processing, and backend functionality.

*Visual Studio Code:

A lightweight code editor used for writing, testing, and debugging Python applications efficiently.

<https://www.code.visualstudio.com/>

*Pandas:

A data analysis library used to organize, clean, and process resume datasets. <https://www.pandas.pydata.org/>

*NumPy:

Provides mathematical and numerical computing support for handling large amounts of data. <https://www.numpy.org/>

*Scikit-learn:

Machine learning framework used for skill prediction, resume classification, and recommendation systems. <https://www.scikit-learn.org/>

*Natural Language Processing (NLP):

Used to extract skills, keywords, and important information from uploaded resumes. <https://www.nltk.org/>

*Flask:

A web framework used to create the user interface and deploy the application locally or on cloud platforms. <https://www.flask.palletsprojects.com/>

*Matplotlib:

Used to generate visual reports and graphical insights from resume analysis results. <https://www.matplotlib.org/>

*Anaconda Navigator:

A desktop graphical user interface used to launch applications and manage python packages and environments for data science. <https://www.anaconda.com/>

*PyCharm:

An Integrated Development Environment (IDE) specifically designed for Python programming, offering code analysis and debugging tools. <https://www.pycharm.com/>

*Git & GitHub:

Version control tools used for project management, collaboration, and source code hosting. <https://github.com/>

*Render:

Cloud deployment platform used to host the web application and make it accessible online. <https://www.render.org/>

System Requirements

- Operating System: Windows 10/11, Linux, or macOS
- Python Version: 3.10 or above
- RAM: Minimum 4 GB

- **Internet Connection:** Required for deployment and package installation
- **Web Browser:** Chrome, Edge, or Firefox